

Assignment #3

22L-6612 BSCS-7G

Operation Research.

Question no # 1: Maximize $z = 12x_1 + 18x_2$
Subject to:

$$5x_1 + 3x_2 \leq 30 \text{ --- (i)}$$

$$2x_1 + 4x_2 \leq 24 \text{ --- (ii)}$$

$$2x_1 + 3x_2 \leq 20 \text{ --- (iii)}$$

$x_1, x_2 \geq 0$ and integers.

From eq (i)

$$5x_1 + 3x_2 = 30$$

$$(0, 10), (6, 0)$$

From eq (iii)

$$2x_1 + 3x_2 = 20$$

$$(0, 20/3), (10, 0)$$

From eq (ii)

$$2x_1 + 4x_2 = 24$$

$$(0, 6), (12, 0)$$

Corner points $\Rightarrow A(0,0), B(0,6), C(?,?), D(6,0)$

$$10x_1 + 6x_2 = 60$$

$$\pm 10x_1 + 20 = \pm 120$$

$$-14x_2 = -60$$

$$x_2 = \frac{30}{7}, x_1 = \frac{60 - 6(30/7)}{10}$$

$$x_1 = \frac{24}{7}$$

$$x_1 = 3.43, x_2 = 4.29$$

$$Z(A) = 0, Z(B) = 108, Z(C) = 118.38, Z(D) = 72$$

So, $x_1 = 3.43, x_2 = 4.24$ is chosen.

we take x_1 and split the problem into two subproblems.

$$x_1 = 3.43, x_1 \leq 3, x_1 \geq 4$$

Subproblem 1

Maximize

$$z = 12x_1 + 18x_2$$

$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \leq 3$$

$x_1, x_2 \geq 0$ and integer

Subproblem 2

Maximize

$$z = 12x_1 + 18x_2$$

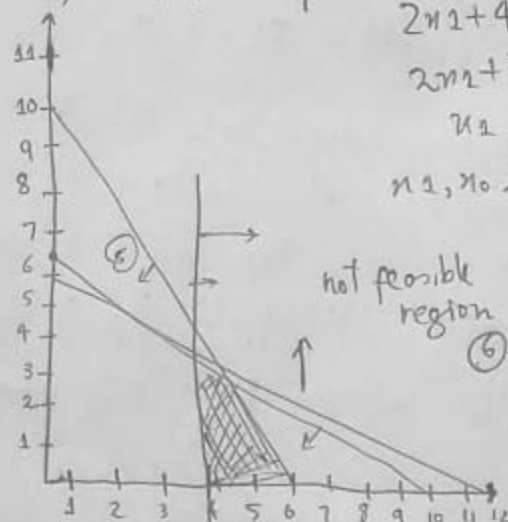
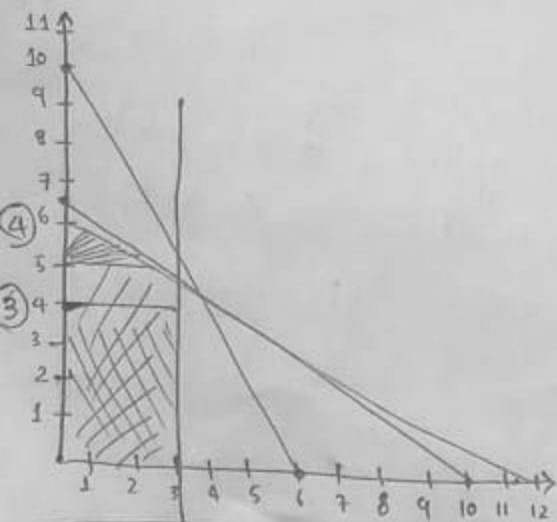
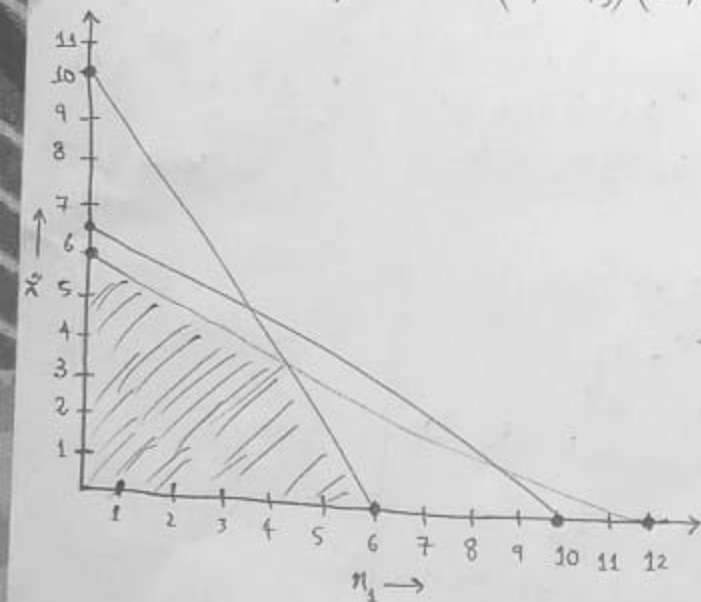
$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \geq 4$$

$x_1, x_2 \geq 0$ and integer



corner points $A(0,0)$, $B(0,6)$, $C(?,?)$, $D(3,0)$
 using eq (iii) and (iv) we cal point C
 $x_1 = 3$

$$2(3) + 4(x_2) = 24 \Rightarrow x_2 = 9/2$$

$$Z(A) = 0, Z(B) = 108, Z(C) = 117$$

$$Z(D) = 36$$

from $Z(C) = 117$, $x_1 = 3$,

$$x_2 = 9/2 = 4.5$$

So, subproblem 1 into 3 and 4

$$x_2 \leq 4, x_2 \geq 5$$

Subproblem 3

$$\text{Max } z = 2x_1 + 18x_2$$

$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \leq 3$$

$$x_2 \leq 4$$

$x_1, x_2 \geq 0$ and
 integers.

corner point $A(0,0)$, $B(0,4)$
 $C(3,4)$, $D(3,0)$

$$Z(A) = 0, Z(B) = 72$$

$$Z(C) = 108, Z(D) = 36$$

So, $Z(C) = 108$

$$x_1 = 3, x_2 = 4$$

Subproblem 4

$$\text{Max } z = 12x_1 + 18x_2$$

$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \leq 3$$

$$x_2 \geq 5$$

$x_1, x_2 \geq 0$ and
 integers

corner points $A(0,5)$,
 $B(0,6)$, $C(3,2)$
 from eq (5) and (2)

$$x_2 = 5 \Rightarrow x_1 = 2$$

$$Z(A) = 90, Z(B) = 108$$

$$Z(C) = 114$$

So, $Z(C) = 114$,
 $x_1 = 2, x_2 = 5$

corner points are $A(4,0)$, $B(?,?)$, $C(6,0)$

$$x_1 = 4, 5(4) + 3x_2 = 30$$

$$x_2 = \frac{30 - 20}{3} = 10/3$$

$$Z(A) = 48, Z(B) = 108, Z(C) = 72$$

So, $Z(B) = 108$, $x_1 = 4$, $x_2 = 3.33$

So, divide subproblem 2 into 5, 6

$$x_2 \leq 3, x_2 \geq 4$$

Subproblem 5

Subproblem 5

$$\text{Max } z = 12x_1 + 18x_2$$

$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \geq 4, x_2 \leq 3$$

$A(4,0)$, $B(4,3)$,
 $C(6,0)$

$$Z(A) = 48$$

$$Z(C) = 72, Z(B) = 102$$

So, $Z(B) = 102$

$$x_1 = 2, x_2 = 3$$

So,

Subproblem 6

$$\text{Max } z = 12x_1 + 18x_2$$

$$5x_1 + 3x_2 \leq 30$$

$$2x_1 + 4x_2 \leq 24$$

$$2x_1 + 3x_2 \leq 20$$

$$x_1 \geq 4$$

$$x_2 \geq 4$$

infeasible

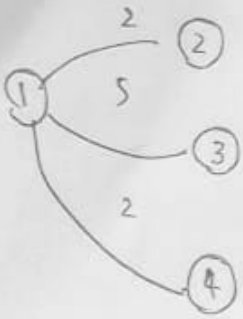
As of maximization,

$$z = 114, x_1 = 2, x_2 = 5$$

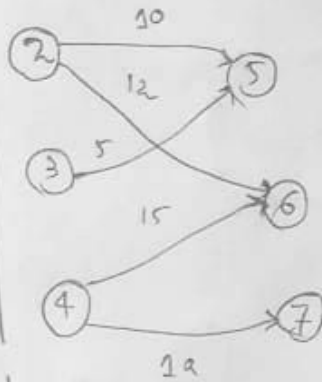
LP relaxation gives a higher value ($z \approx 118.29$) because it allows
 fraction values of x_1 and x_2 . when integrity is followed in the solution
 must be whole numbers so, the objective value drops to $z = 114$.
 This difference is called integrity gap. and shows, LP relaxation
 provides upper bound for integer problems.

Question no #3

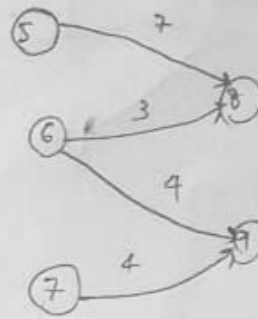
f₁



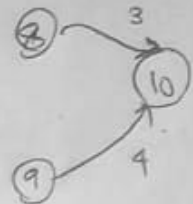
f₂



f₃



f₄



Using DP backward Recursive Algo,

Stage 4:-

x ₄	10	f ₄ (x ₄)	min distance
8	3	3	10
9	4	4	10

Stage 3:-

x ₃	8	9	f ₃ (x ₃)	min distance
5	7+3=10	-	10	8
6	3+3=6	8	6	8
7	-	8	8	9

Stage 2:-

x ₂	5	6	7	f ₂ (x ₂)	min distance
2	10+10=20	18	-	18	6
3	5+10=15	-	-	15	5
4	-	21	27	21	6

x ₁	2	3	4	f ₁ (x ₁)	min distance
1	2+18=20	5+15=20	21+21=23	20	2 or 3

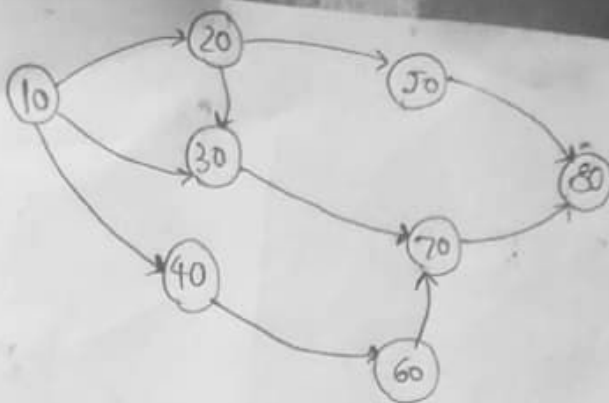
minimum distance is 20km from 10 to 1.

paths:-

1 ← 2 ← 6 ← 8 ← 10 ⇒ 20km

1 ← 3 ← 5 ← 7 ← 8 ← 10
⇒ 20km

Question no # 1



$$\frac{a+4m+b}{6}$$

$$\left(\frac{b-a}{6}\right)^2$$

expected time | variance

11.67 $\Rightarrow$ 12 | 4

10.16 $\Rightarrow$ 10 | 0.70

10.8 $\Rightarrow$ 11 | 0.25

7.67 $\Rightarrow$ 8 | 0.44

11 | 0.44

18 | 1.78

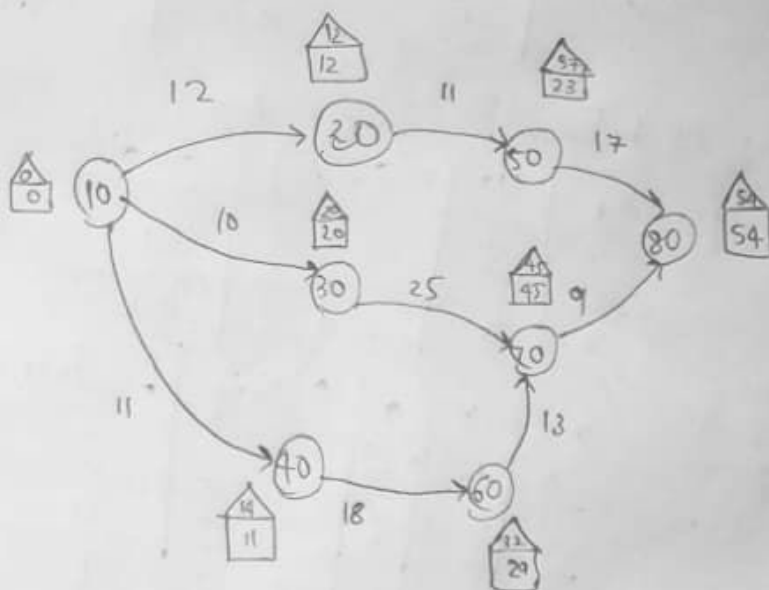
25.16 $\Rightarrow$ 25 | 2.25

12.8 $\Rightarrow$ 13 | 2.25

17.16 $\Rightarrow$ 17 | 1.36

9

1



critical path $\rightarrow$ (10) $\rightarrow$ (20) $\rightarrow$ (30) $\rightarrow$ (70) $\rightarrow$ (80)

expected project completion time = 12 + 8 + 25 + 9 $\Rightarrow$ 54 days

Variance of project = 4 + 0.44 + 2.25 + 1 = 7.67

SD of = $\sqrt{7.69} = 2.77$

total float	slack
10	20
10	30
10	40
20	30
20	50
40	60
30	70
60	70
50	80
70	80

activities with zero delay are 10-20, 20-30, 30-70, 70-80.
nodes with zero delay = 10, 20, 30, 70, 80

$$z \text{ score} = \frac{\text{Target} - \text{Expectation}}{\text{SD}} = \frac{50 - 54}{2.77} = -1.44$$

using table value = 0.0749 with means

7.49%
chances to be completed within 50 days.